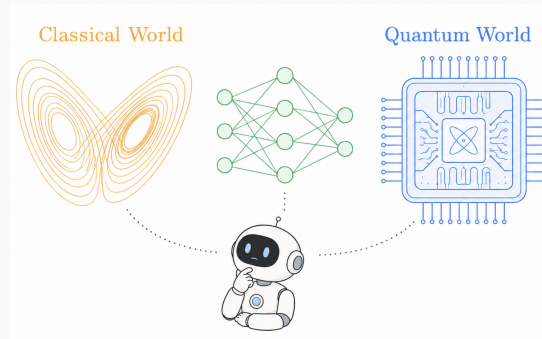


AI Design of Quantum Processors

🎯 Research Project Goal

Overarching Goal: Predict properties of quantum processors using classical data. The project proceeds through three research components:



⚙️ Component 1

Classical Data Generation

Simulate classical systems with Newton's equations.

⚙️ Component 2

Quantum Data Generation

Calculate quantum properties with QuTiP and squbits.

🧠 Component 3

ML Training

Train and validate an ML model on the generated data.

☰ Task Status Overview

✓ = accomplished tasks **New** = newly added tasks

R1: Classical Data

- ✓ Task 1: Energy & Phase Space
- ✓ Task 2: Dynamics & Trajectories
- New** Task 3: Cosine Potential
- New** Task 4: Two Coupled Oscillators

R2: Quantum Data

- ✓ Task 1: Operators & Spectrum
- ✓ Task 2: State Dynamics & Wigner
- New** Task 3: Fluxonium Dynamics

R3: ML Training

- New** Task 1: Classical-to-Quantum Regression

📖 References & Resources

- **PI:** Feel free to reach out with any questions you may have.
- **QuTiP** (quantum toolbox in Python): qutip.org
- **scqubits** (superconducting qubit library): scqubits.readthedocs.io
- **PyTorch** (machine learning framework): pytorch.org
- **Quantum Mechanics** (Essler, Oxford): [lecture notes PDF](#)
- **Superconducting quantum processors** (basic overview): [PennyLane tutorial](#)

★ Best Practices

🔗 Coding

- **Write readable code.** Use descriptive variable names, break work into short functions, and add a one-line comment for every non-obvious step. All code should be written in **Jupyter notebooks**.
- **Use a shared GitHub repository.** All your notebooks and scripts must live in the group GitHub repo. Push your changes regularly and keep the PI added as a collaborator so progress is always visible.

📊 Plots

- **Label everything.** Every axis must carry a label with units (or state dimensionless). Every curve needs a legend entry. Every figure needs a caption that explains what is shown and what the key takeaway is.
- **Choose colors intentionally.** Use perceptually uniform colormaps (e.g. `viridis`, `plasma`) so that differences in data are faithfully represented on screen and in print.
- **Size fonts for the medium.** Set `fontsize` so that tick labels, axis titles, and legends remain legible once the figure is scaled to a slide or a printed page.

👤 Meeting Slides

- **Organize your latest calculations into slides.** For each meeting, collect your most recent results as polished plots in a slide deck. Annotate the key features directly on the figures rather than describing them verbally.
- **One idea per slide.** State the main takeaway in the slide title as a complete sentence so the audience understands your point even if they only skim the titles.
- **Structure each update in three steps:**
 1. *Context:* discuss why you were doing this calculation and what question it was meant to answer.
 2. *Results:* present what you found, supported by figures.
 3. *Open questions:* describe what you do not yet understand and what you would like to do or discuss next.

Research Component 1: Classical Data Generation

✓ Task 1: Classical Harmonic Oscillator: Energy and Phase Space

Coding

The harmonic oscillator is one of the most important models in physics. It describes small oscillations near a stable equilibrium and appears in mechanics, circuits, vibrations, and quantum hardware. In this task you will study the classical energy function and its phase-space structure.

- (a) State the classical energy $E(x, p)$ of a one-dimensional harmonic oscillator in terms of position x , momentum p , mass m , and angular frequency ω .
- (b) The equations of motion can be derived from the energy by treating it as the Hamiltonian $H(x, p)$. Starting from Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial x},$$

derive the classical equations of motion for $x(t)$ and $p(t)$. Combine them to obtain the familiar second-order equation for $x(t)$.

- (c) Make a plot of the classical energy $E(x, p)$ in phase space. Use either a contour plot or a density plot over a two-dimensional grid of (x, p) values. Explain the shape of the constant-energy curves.

✓ Task 2: Classical Harmonic Oscillator: Dynamics and Trajectories

Coding

The energy function tells us which phase-space curves are allowed; dynamics tells us how the system moves along those curves. In this task you will simulate and visualize classical harmonic oscillator trajectories.

- (a) Solve the classical equations of motion derived in Task 1. Choose one initial condition (x_0, p_0) and make a phase-space plot of the trajectory $(x(t), p(t))$. Indicate the initial point on the plot.
- (b) Repeat the simulation for a random set of initial conditions. Plot all trajectories on the same axes and explain how the trajectory shapes depend on the initial energy.

New Task 3: Nonlinear Oscillator: Cosine Potential and Phase Space

Coding

In this task you will extend the single-oscillator model by adding a cosine term to the potential.

Consider the Hamiltonian

$$H(x, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - V_0 \cos(kx),$$

where V_0 and k are constants that control the strength and spatial period of the nonlinearity.

- (a) Starting from Hamilton's equations, write the equations of motion for $x(t)$ and $p(t)$. Identify the new term that arises from the cosine potential and explain how it modifies the restoring force compared to the pure harmonic case.
- (b) For several values of the initial energy $E_0 = H(x_0, p_0)$, simulate the dynamics and plot the phase-space trajectory $(x(t), p(t))$ on the same axes.
- (c) Choose a reference energy E_{ref} and generate a set of initial conditions whose energies lie in the interval $[E_{\text{ref}} - \Delta E, E_{\text{ref}} + \Delta E]$. Plot all resulting trajectories in phase space on the same figure. Use color or transparency to encode the initial energy within this band.

New Task 4: Two Coupled Oscillators: Momentum Coupling and Poincaré Maps Coding

In this task you will study a pair of oscillators with a cosine nonlinearity and momentum-momentum coupling.

Consider the Hamiltonian

$$H = \sum_{i=1}^2 \left(\frac{p_i^2}{2m} + \frac{1}{2} m \omega_i^2 x_i^2 - V_0 \cos(kx_i) \right) + \lambda p_1 p_2,$$

where λ controls the strength of the $\mathbf{p}_1 \cdot \mathbf{p}_2$ coupling between the two oscillators.

- (a) Derive the equations of motion for (x_1, p_1) and (x_2, p_2) .
- (b) The full phase space is four-dimensional. For a fixed initial energy E_0 , plot the four natural two-dimensional projections:
 - i. (x_1, p_1) ,
 - ii. (x_2, p_2) ,
 - iii. (x_1, x_2) ,
 - iv. (p_1, p_2) .

Repeat for several values of E_0 and describe how the structure of each projection changes with energy.

- (c) Choose a reference energy E_{ref} and simulate trajectories for initial conditions with energies in $[E_{\text{ref}} - \Delta E, E_{\text{ref}} + \Delta E]$. Overlay the (x_1, p_1) projections on one figure and the (x_2, p_2) projections on another, encoding initial energy by color.
- (d) Construct Poincaré maps for the two-oscillator system. Choose a suitable surface of section (for example, record points where $x_2 = 0$ with $p_2 > 0$) and plot the map for at least two distinct energy values. Discuss whether the maps reveal regular islands, quasiperiodic orbits, or chaotic scattering.

Research Component 2: Quantum Data Generation

✓ Task 1: Quantum Harmonic Oscillator: Operators and Energy Spectrum

Conceptual Coding

In quantum mechanics, physical observables are represented by operators and the energy spectrum is discrete. In this task you will construct the quantum Hamiltonian and compute its eigenvalues.

- (a) In quantum mechanics, position and momentum become the operators $\hat{x}$ and $\hat{p}$. Using QuTiP, construct the truncated matrix representations of

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger), \quad \hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a} - \hat{a}^\dagger),$$

and

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2.$$

Make `imshow` plots of the absolute values of the matrix elements of $\hat{x}$, $\hat{p}$, and $\hat{H}$.

- (b) The possible energies of a quantum system are the eigenvalues of the Hamiltonian. Use QuTiP to calculate the eigenvalues of $\hat{H}$. Plot the first several eigenvalues versus the level index n , and compare your numerical values to the analytic expression

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

- (c) Briefly explain what changed, and what stayed the same, when moving from the classical energy function $E(x, p)$ to the quantum Hamiltonian matrix $\hat{H}$.

✓ Task 2: Quantum Harmonic Oscillator: State Dynamics and Phase Space

Conceptual Coding

Quantum dynamics are governed by the Schrödinger equation, and quantum states can be visualized in phase space using the Wigner function. In this task you will simulate quantum time evolution and compare it to classical motion.

- (a) Use QuTiP's `sesolve` function to solve the Schrödinger equation,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

for the following initial states:

- i. an energy eigenstate $|n\rangle$, such as $|0\rangle$ or $|1\rangle$,
- ii. the normalized superposition $(|0\rangle + |1\rangle)/\sqrt{2}$,
- iii. a displaced Gaussian state, implemented as a coherent state $|\alpha\rangle$ or as $\hat{D}(\alpha)|0\rangle$.

Store the quantum state at each time step so that you can visualize its dynamics.

- (b) The Wigner function $W(x, p)$ is a phase-space representation of a quantum state. It allows us to compare quantum states to classical motion in the same (x, p) plane. For each initial state in part (a), use QuTiP to compute the Wigner function over a grid of x and p values. Make a static plot of $W(x, p)$ at $t = 0$.
- (c) Make a movie of the Wigner function dynamics for each of the three initial states from part (a). Use a fixed color scale for each movie so that changes in the shape and sign of $W(x, p)$ are visually meaningful.
- (d) For each quantum trajectory, calculate the expectation values $\langle \hat{x} \rangle(t)$ and $\langle \hat{p} \rangle(t)$. Plot $\langle \hat{p} \rangle$ versus $\langle \hat{x} \rangle$ in phase space. On the same axes, overlay the classical solution with initial condition

$$x_0 = \langle \hat{x} \rangle(0), \quad p_0 = \langle \hat{p} \rangle(0).$$

Discuss when the expectation-value motion agrees with the classical trajectory and what information is lost when only the averages are plotted.

- (e) Write a short reflection, no more than one page, explaining what each representation tells you: energy contours, matrix elements, eigenvalues, Wigner functions, and expectation-value phase-space plots.

New Task 3: Fluxonium: Wave Packets and Phase-Charge Dynamics

Conceptual Coding

A fluxonium qubit is a superconducting circuit in which a Josephson junction is shunted by a large inductor. The circuit has charging energy, inductive energy, and Josephson nonlinearity. Its dynamics are governed by conjugate phase and charge variables.

In this task you will build the fluxonium Hamiltonian, construct localized wave packets, simulate quantum time evolution, and compare the results to the corresponding classical motion.

The quantum Hamiltonian is

$$\hat{H} = 4E_C \hat{n}^2 + \frac{1}{2} E_L (\hat{\phi} - \phi_{\text{ext}})^2 - E_J \cos \hat{\phi},$$

where $E_C = e^2/(2C_\Sigma)$ is the charging energy, $E_L = \Phi_0^2/L$ is the inductive energy, E_J is the Josephson energy, and ϕ_{ext} is the reduced external flux threading the loop.

This Hamiltonian is structurally similar to the classical nonlinear oscillator from Task 3,

$$H(x, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 - V_0 \cos(kx).$$

The correspondence between the two models is

$$x \leftrightarrow \phi - \phi_{\text{ext}}, \quad p \leftrightarrow n, \quad V_0 \leftrightarrow E_J, \quad k = 1.$$

Matching the kinetic terms, $p^2/(2m) \leftrightarrow 4E_C \hat{n}^2$, gives $m = 1/(8E_C)$ in units where $\hbar = 1$. Matching the harmonic terms, $\frac{1}{2} m \omega^2 x^2 \leftrightarrow \frac{1}{2} E_L (\hat{\phi} - \phi_{\text{ext}})^2$, gives $\omega = \sqrt{8E_C E_L}$.

External flux ϕ_{ext} shifts the cosine potential through $x = \phi - \phi_{\text{ext}}$. Use classical initial conditions (ϕ_0, n_0) matched to the centers of the quantum wave packets.

Use **scqubits** to construct the fluxonium Hamiltonian and obtain its spectrum. Use **QuTiP** for time evolution. Suggested starting parameters are $E_J/E_C \approx 5$, $E_L/E_C \approx 0.5$, $\phi_{\text{ext}} = \pi$ (half-flux sweet spot), and a Hilbert-space truncation of ~ 40 – 60 charge states.

- (a) State the fluxonium Hamiltonian $\hat{H}$ explicitly and explain its similarity to the classical Hamiltonian from Task 3. Identify which terms provide the kinetic (charging) energy, the harmonic (inductive) energy, and the nonlinear (Josephson) potential.
- (b) Use **scqubits** to compute the first four eigenstates of $\hat{H}$. Plot their energies and wavefunctions in the phase/charge basis.
- (c) Construct a normalized wave packet localized and centered at a phase value ϕ_0 (and optionally displaced in charge by n_0), for example a Gaussian in ϕ projected onto the truncated Hilbert space. Make a grid plot of the packet at $t = 0$ (phase versus probability density, or a Wigner-style representation).
- (d) Use QuTiP's **sesolve** with this wave packet as the initial state to solve the Schrödinger equation. Calculate the expectation values $\langle \hat{\phi} \rangle(t)$, $\langle \hat{n} \rangle(t)$, and the mean energy $\langle \hat{H} \rangle(t)$.
- (e) With $x_0 = \phi_0 - \phi_{\text{ext}}$ and $p_0 = n_0$, compute the classical energy $E_0 = H(x_0, p_0)$ using the mapped parameters above and overlay the classical trajectory $(x(t), p(t))$ against $\langle \hat{\phi} \rangle(t)$, $\langle \hat{n} \rangle(t)$ in phase space. Discuss where the quantum and classical motions agree and where they differ.
- (f) Repeat parts (c)–(e) for several values of ϕ_0 (e.g. $\phi_0 \in \{-\pi, -\pi/2, 0, \pi/2, \pi\}$, with fixed or varied n_0). Present the results as grid plots showing the initial packet, expectation-value trajectories, and classical overlays.

Research Component 3: ML Training

New Task 1: Fluxonium: Classical-to-Quantum Regression

Coding

The overarching goal of this project is to predict quantum dynamics from classical data. In this task you will generate paired classical and quantum trajectory datasets for the fluxonium model and train a neural network to learn the map between them.

Use the **same** mapped parameters $(E_J, E_C, E_L, \phi_{\text{ext}}, m, \omega, V_0, k)$ for both the quantum Hamiltonian from Task 3 and the classical Hamiltonian from Task 3,

$$H(x, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - V_0 \cos(kx),$$

with $x \leftrightarrow \phi - \phi_{\text{ext}}$, $p \leftrightarrow n$, $V_0 \leftrightarrow E_J$, $k = 1$, $m = 1/(8E_C)$, and $\omega = \sqrt{8E_C E_L}$.

- (a) Implement a shared parameter module so that classical and quantum simulations use consistent values. For each sample, draw a random initial packet centered at (ϕ_0, n_0) ; the quantum simulation uses this packet as its initial state, and the classical simulation uses $(x_0, p_0) = (\phi_0 - \phi_{\text{ext}}, n_0)$.
- (b) **Quantum dataset.** Following Task 3, generate N_s quantum trajectories. Each sample $B_i \in \mathbb{R}^{2N_t}$ is the vector

$$B_i = (\langle x \rangle(t_1), \dots, \langle x \rangle(t_{N_t}), \langle p \rangle(t_1), \dots, \langle p \rangle(t_{N_t})),$$

where $x \equiv \phi$ and $p \equiv n$, and N_t is the number of time grid points. Choose the final time t_{final} so that at least one approximately full cyclic evolution is completed (estimate from the classical period or from oscillations of $\langle \hat{H} \rangle$).

- (c) **Classical dataset.** Using the classical Hamiltonian above, generate N_s paired trajectories with the **same** random (ϕ_0, n_0) initial conditions as in part (b). Each sample $A_i \in \mathbb{R}^{2N_t}$ is the vector

$$A_i = (x(t_1), \dots, x(t_{N_t}), p(t_1), \dots, p(t_{N_t}))$$

on the same time grid.

- (d) **PyTorch regression model.** Implement a supervised vector-to-vector regression model that learns a map $f_\theta : A_i \mapsto \hat{B}_i$ from classical input $A_i \in \mathbb{R}^{d_A}$ to quantum output $B_i \in \mathbb{R}^{d_B}$, where $d_A = d_B = 2N_t$. Use a fully connected multilayer perceptron with two hidden layers and ReLU activations:

$$h_1 = \text{ReLU}(W_1 A + b_1), \quad h_2 = \text{ReLU}(W_2 h_1 + b_2), \quad \hat{B} = W_3 h_2 + b_3,$$

where $W_1 \in \mathbb{R}^{d_1 \times d_A}$, $W_2 \in \mathbb{R}^{d_2 \times d_1}$, and $W_3 \in \mathbb{R}^{d_B \times d_2}$.

Train with mean squared error as the loss, the Adam optimizer, an 80%/20% train/validation split, and mini-batch training.

Monitor both training and validation loss after each epoch. Explore the hidden-layer widths d_1 and d_2 together with the learning rate, mini-batch size, and number of epochs to assess generalization. Report validation mean squared error, and optionally relative or component-wise prediction errors.